

		Elastic means relative velocity of approach = relative velocity of separation By relative velocity of approach and separation, KE not conserved for option B.
5	A	By conservation of energy, elastic potential energy is converted to kinetic energy and work done against friction. As height is the same at initial and final positions, gravitational potential energy is unchanged. $EPE_{loss} = KE_{gain} + WD_{friction}$ $\frac{1}{2}kx^2 = (KE_{final} - KE_{initial}) + fd$ $KE_{final} = \frac{1}{2}kx^2 - fd \quad (\text{since } KE_{initial} = 0)$

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D

m s^{-2}